

Topology & Multivariable Calculus

FMA_2F002_EP

Lecture Notes

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These are personal lecture notes and may contain errors. For the official course material consult the lecturer.

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1 Norms in Vector Spaces

1.1 Setting

Let $E, F, G, \dots$ be vector spaces over $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$.

- If $\mathbb{K} = \mathbb{R}$, for $x \in \mathbb{R}$, $|x| = \max(x, -x)$.
- If $\mathbb{K} = \mathbb{C}$, for $z = x + iy$, we have $|z| = \sqrt{x^2 + y^2} = l$. In polar form, $z = le^{i\theta}$ where $l \geq 0$ and $\theta \in [0, 2\pi)$.

1.2 Norms

Definition 1: Norm

A norm on E is a map $N : E \rightarrow \mathbb{R}_{\geq 0} = \mathbb{R}^+ = [0, +\infty)$ satisfying the following 3 properties:

1. **Triangular inequality:** $\forall x, y \in E, N(x + y) \leq N(x) + N(y)$.
2. **Absolute homogeneity:** $\forall \lambda \in \mathbb{K}, \forall x \in E, N(\lambda x) = |\lambda|N(x)$.
3. **Positive definiteness:** $N(x) = 0 \Leftrightarrow x = 0$.

Remark 2

Positive definiteness is equivalent to the fact that N "separates points". That is, $\forall x, y \in E$, $N(x - y) = 0$ if and only if $x = y$.

Proposition 3: Reverse Triangular Inequality

The triangular inequality is equivalent to the reverse triangular inequality: $\forall x, y \in E$, $|N(x) - N(y)| \leq N(x - y)$.

PROOF. Let $x, y \in E$. Then $N(x) = N(x - y + y) \leq N(x - y) + N(y)$, which implies $N(x) - N(y) \leq N(x - y)$. Similarly, $N(y) - N(x) \leq N(y - x)$. Since $N(x - y) = |-1|N(y - x)$, we obtain $|N(x) - N(y)| \leq N(x - y)$. $\square$

Example 4: Common Norms

- $|\cdot|$ is a norm on $\mathbb{R}$.
- On $E = \mathbb{R}^d$, Sup Norm $\|x\|_{\infty} = \max_{i=1}^d |x_i|$.
- Euclidean norm on $\mathbb{R}^d$: $\|x\|_2 = \sqrt{\sum_{i=1}^d x_i^2}$.
- On $F = C^0([0, 1], \mathbb{R})$, Sup Norm $\|f\|_{\infty} = \sup_{x \in [0, 1]} |f(x)|$.
- On F , L_1 norm $\|f\|_{L^1} = \int_0^1 |f(x)| dx$.
- On F , L_2 norm $\|f\|_{L^2} = \sqrt{\int_0^1 |f(x)|^2 dx}$.

Lemma 5: Norms on $\mathbb{R}$

Let $N : \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$ be a norm. Then $\exists \alpha > 0$ such that $N(\cdot) = \alpha |\cdot|$.

PROOF. Let $x \in \mathbb{R}$. Then $N(x) = N(x \cdot 1) = |x|N(1)$. Let $\alpha = N(1) > 0$. $\square$

Lemma 6: Euclidean Norm on $\mathbb{R}^d$

$\|\cdot\|_2$ is a norm on $E = \mathbb{R}^d$.

PROOF. Absolute homogeneity is obvious. For positive definiteness: $\|x\|_2 = 0 \Rightarrow \sum_{i=1}^d x_i^2 = 0 \Rightarrow x_i = 0$ for all $i = 1, \dots, d \Rightarrow x = 0_E$. For the triangular inequality, we use the Euclidean scalar product $\langle x, y \rangle = \sum_{i=1}^d x_i y_i$. This product is symmetric, $\mathbb{R}$ -bilinear, and $\langle x, x \rangle = 0 \Rightarrow x = 0_E$. We have $\|x + y\|_2 = \sqrt{\langle x + y, x + y \rangle} = \sqrt{\|x\|_2^2 + \|y\|_2^2 + 2\langle x, y \rangle}$. By the Cauchy-Schwarz inequality, $|\langle x, y \rangle| \leq \|x\|_2 \|y\|_2$. Thus, $\|x + y\|_2 \leq \sqrt{\|x\|_2^2 + 2\|x\|_2 \|y\|_2 + \|y\|_2^2} = \|x\|_2 + \|y\|_2$. $\square$

Lemma 7: Sup Norm on $C^0([0, 1], \mathbb{R})$

$\|\cdot\|_\infty$ is a norm on $F = C^0([0, 1], \mathbb{R})$.

PROOF. For any $f \in F$, f is bounded. Positive homogeneity: $\|\lambda f\|_\infty = \sup |\lambda f(x)| = |\lambda| \|f\|_\infty$. Positive definiteness: If $\|f\|_\infty = 0$, then $0 \leq |f(x)| \leq \sup |f(x)| = 0$, which implies $f(x) = 0$ for all x , so $f = 0_F$. Triangular inequality: $|(f + g)(x)| = |f(x) + g(x)| \leq |f(x)| + |g(x)| \leq \|f\|_\infty + \|g\|_\infty$. Taking the supremum yields the result. $\square$

1.3 Distances

Definition 8: Distance

A distance on a set $X \neq \emptyset$ is a map $d : X^2 \rightarrow \mathbb{R}^+$ satisfying:

1. **Triangular inequality:** $\forall x, y, z \in X, d(x, z) \leq d(x, y) + d(y, z)$.
2. **Symmetry:** $\forall x, y \in X, d(x, y) = d(y, x)$.
3. **Positive definiteness:** $\forall x, y \in X, d(x, y) = 0 \Leftrightarrow x = y$.

Example 9: Discrete Distance

Let $X \neq \emptyset$. The discrete distance is defined as $d(x, y) = 0$ if $x = y$, and $d(x, y) = 1$ if $x \neq y$.

Proposition 10: Distance Induced by a Norm

Let (E, N) be a normed vector space (NVS) over $\mathbb{K}$. The map $d_N : E^2 \rightarrow \mathbb{R}^+$ given by $d(x, y) = N(x - y)$ is a distance on E .

PROOF. Symmetry: $d(x, y) = N(x - y) = N(y - x) = d(y, x)$. Positive definiteness: $d(x, y) = 0 \Leftrightarrow N(x - y) = 0 \Rightarrow x - y = 0 \Rightarrow x = y$. Triangular inequality: $d(x, z) = N(x - z) =$

$$N(x - y + y - z) \leq N(x - y) + N(y - z) = d(x, y) + d(y, z). \quad \square$$

1.4 Balls and Convexity

Definition 11: Balls

Let (X, d) be a metric space. For $x_0 \in X$ and $r \in \mathbb{R}^+$, we define:

- **Open ball:** $B(x_0, r) = \{x \in X \mid d(x_0, x) < r\}$.
- **Closed ball:** $\overline{B}(x_0, r) = \{x \in X \mid d(x_0, x) \leq r\}$.

In E , $B(0, 1)$ is defined as the unit ball.

Definition 12: Segments and Convexity

For $x, y \in E$, the closed segment is $[x, y] = \{tx + (1 - t)y \mid t \in [0, 1]\}$. The open segment is $(x, y) = [x, y] \setminus \{x, y\}$. A subset $C \subseteq E$ is **convex** if for all $x, y \in C$, the segment $[x, y] \subseteq C$.

Proposition 13: Convex Subsets

1. Convex subsets of $\mathbb{R}$ are intervals.
2. Balls in a NVS are convex.

PROOF. We prove (2) for $B = B(x_0, r)$ with $r > 0$. Let $x, y \in B$, meaning $N(x - x_0) < r$ and $N(y - x_0) < r$. Let $z = tx + (1 - t)y \in [x, y]$ with $t \in [0, 1]$. We have $d(z, x_0) = N(z - x_0) = N(t(x - x_0) + (1 - t)(y - x_0))$. By the triangular inequality, $d(z, x_0) \leq tN(x - x_0) + (1 - t)N(y - x_0) < tr + (1 - t)r = r$. Thus, $z \in B$, which implies $[x, y] \subseteq B$, meaning B is convex. $\square$

2 Sequences

Definition 1: Convergence of Sequences

Let $\underline{x} = (x_n)_{n \in \mathbb{N}}$ be a sequence in a $\mathbb{K}$ -NVS E . We say $x_n \rightarrow x_\infty$ in E (wrt N) if $N(x_n - x_\infty) \rightarrow 0$ in $\mathbb{R}$. We denote this as $x_n \rightarrow x_\infty$ or $\lim x_n = x_\infty$. If a sequence does not converge, it diverges.

Example 2: Convergence in Different Norms

Consider $E = C^0([0, 1], \mathbb{R})$ and the sequence of functions $f_n(x) = x^n$.

- For the sup norm $\|\cdot\|_\infty$, we have $\|f_n\|_\infty = 1$ for all n , so it does not converge to 0_E .
- For the L^1 norm, we have $\|f_n\|_{L^1} = \int_0^1 x^n dx = \frac{1}{n+1} \rightarrow 0$. Thus $f_n \rightarrow 0_E$ for $\|\cdot\|_{L^1}$.

Proposition 3: Uniqueness of the Limit

The limit of a convergent sequence is unique.

PROOF. Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in E converging to both $x_\infty \in E$ and $x'_\infty \in E$. By the triangular inequality, $0 \leq N(x_\infty - x'_\infty) \leq N(x_\infty - x_n) + N(x_n - x'_\infty)$. Since $x_n \rightarrow x_\infty$ and $x_n \rightarrow x'_\infty$, both $N(x_\infty - x_n) \rightarrow 0$ and $N(x_n - x'_\infty) \rightarrow 0$ as $n \rightarrow \infty$. By the squeeze theorem, $N(x_\infty - x'_\infty) = 0$, which implies $x_\infty = x'_\infty$. $\square$

Proposition 4: Continuity of the Norm

1. If $x_n \rightarrow x_\infty$ in E , then $N(x_n) \rightarrow N(x_\infty)$ in $\mathbb{R}$.
2. Hence, $(N(x_n))_{n \in \mathbb{N}}$ is a bounded sequence, and $N(x_\infty) \leq \sup_{n \in \mathbb{N}} N(x_n)$.

PROOF. By the reverse triangular inequality, $0 \leq |N(x_n) - N(x_\infty)| \leq N(x_n - x_\infty)$. Since $x_n \rightarrow x_\infty$, we have $N(x_n - x_\infty) \rightarrow 0$. By the squeeze theorem, $|N(x_n) - N(x_\infty)| \rightarrow 0$, meaning $N(x_n) \rightarrow N(x_\infty)$. The second part follows from the fact that any convergent sequence $(u_n)_{n \in \mathbb{N}}$ in $\mathbb{R}$ is bounded (i.e., $\sup_{n \in \mathbb{N}} |u_n| < +\infty$). Its limit satisfies $|u_\infty| \leq \sup_{n \in \mathbb{N}} |u_n|$. $\square$

Proposition 5: Algebraic Operations on Limits

Let $(x_n)_{n \in \mathbb{N}}$ and $(y_n)_{n \in \mathbb{N}}$ be sequences in E , and let $(\lambda_n)_{n \in \mathbb{N}}$ be a sequence in $\mathbb{K}$. Assume:

1. $x_n \rightarrow x_\infty$ in E
2. $y_n \rightarrow y_\infty$ in E
3. $\lambda_n \rightarrow \lambda_\infty$ in $\mathbb{K}$

Then, $x_n + y_n \rightarrow x_\infty + y_\infty$ in E , and $\lambda_n y_n \rightarrow \lambda_\infty y_\infty$ in E .

PROOF. For $n \in \mathbb{N}$, set $z_n = x_n + \lambda_n y_n$ and $z_\infty = x_\infty + \lambda_\infty y_\infty$. We want to show $N(z_n - z_\infty) \rightarrow 0$. By the triangular inequality, $0 \leq N(z_n - z_\infty) \leq N(x_n - x_\infty) + N(\lambda_n y_n - \lambda_\infty y_\infty)$. By adding and subtracting $\lambda_\infty y_n$, we get $N(\lambda_n y_n - \lambda_\infty y_\infty) \leq |\lambda_n - \lambda_\infty| N(y_n) + |\lambda_\infty| N(y_n - y_\infty)$. Since $y_n \rightarrow y_\infty$, the sequence $(N(y_n))_{n \in \mathbb{N}}$ is bounded by some $M > 0$. Thus, $N(\lambda_n y_n - \lambda_\infty y_\infty) \leq M|\lambda_n - \lambda_\infty| + |\lambda_\infty| N(y_n - y_\infty)$. As $n \rightarrow \infty$, $|\lambda_n - \lambda_\infty| \rightarrow 0$ and $N(y_n - y_\infty) \rightarrow 0$, so $N(\lambda_n y_n - \lambda_\infty y_\infty) \rightarrow 0$. Consequently, $N(z_n - z_\infty) \rightarrow 0$, which completes the proof. $\square$

Definition 6: Subsequence and Adherence Value

- A sequence $(y_n)_{n \in \mathbb{N}}$ is a **subsequence** (or extraction) of $(x_n)_{n \in \mathbb{N}}$ if there exists a strictly increasing function $\varphi : \mathbb{N} \rightarrow \mathbb{N}$ such that $y_n = x_{\varphi(n)}$.
- An element $l \in E$ is an **adherence value** (or accumulation point) of $(x_n)_{n \in \mathbb{N}}$ if there exists a strictly increasing extraction $\varphi : \mathbb{N} \rightarrow \mathbb{N}$ such that $x_{\varphi(n)} \rightarrow l$.

Problem 7: Accumulation Points of $\mathbb{Q}$

The set of rational numbers $\mathbb{Q}$ is countable, so it can be written as a sequence $\mathbb{Q} = \{q_0, q_1, \dots\}$. What are the accumulation points of the sequence $(q_n)_{n \in \mathbb{N}}$?

Remark 8

Because $\mathbb{Q}$ is dense in $\mathbb{R}$, every real number is an accumulation point. Thus, the set of accumulation points is all of $\mathbb{R}$.

3 Lecture 2: Open & Closed Subsets in Normed Vector Spaces

Let $(E, \|\cdot\|)$ be a fixed NVS over $\mathbb{K}$.

3.1 Open Subsets

Definition 1: Open Subset

A subset $U \subset E$ is **open** if for all $u \in U$, there exists $\varepsilon > 0$ such that $B(u, \varepsilon) \subset U$.

Example 2: Open Sets

- $\emptyset$ and E are open subsets.
- In $\mathbb{R}$, open intervals (a, b) are open subsets.

Proposition 3: Open Balls are Open Sets

In $(E, \|\cdot\|)$, open balls are open subsets.

PROOF. Let $B = B(x_0, r) \subset E$ with $r > 0$. Let $y \in B$. We have $d(x_0, y) = \|x_0 - y\| = \delta < r$. Choose $\varepsilon_y = r - \delta = r - \|x_0 - y\| > 0$. We claim $B(y, \varepsilon_y) \subset B$. Let $z \in B(y, \varepsilon_y)$. By the triangular inequality: $d(x_0, z) \leq d(x_0, y) + d(y, z) = \|x_0 - y\| + \|y - z\| < \delta + \varepsilon_y = r$. Therefore, $z \in B$, which implies $B(y, \varepsilon_y) \subset B$. Thus, B is open. $\square$

Proposition 4: Union of Open Balls

The open subsets of E are exactly the unions of open balls.

PROOF. Let $U \subset E$ be an open subset. By definition, for every $u \in U$, there exists $\varepsilon_u > 0$ such that $B(u, \varepsilon_u) \subset U$. Then $U \subset \bigcup_{u \in U} B(u, \varepsilon_u) \subset U$. Thus, $U = \bigcup_{u \in U} B(u, \varepsilon_u)$, which is a union of open balls. $\square$

3.2 Closed Subsets

Definition 5: Closed Subset

A subset $F \subset E$ is **closed** if its complement $F^c = E \setminus F$ is open.

Example 6: Closed Sets

$\emptyset$ and E are closed subsets of $(E, \|\cdot\|)$.

Proposition 7: Properties of Open and Closed Sets

1. Arbitrary unions of open subsets are open.
2. Finite intersections of open subsets are open.
3. Arbitrary intersections of closed subsets are closed.

4. Finite unions of closed subsets are closed.

Remark 8

An infinite intersection of open sets is not necessarily open. For example, in $\mathbb{R}$, the intersection $\bigcap_{n \geq 1} (-\frac{1}{n}, \frac{1}{n}) = \{0\}$ is not open.

PROOF. Let us prove the properties for open sets:

1. Let $(O_i)_{i \in I}$ be a family of open subsets. Let $\mathcal{O} = \bigcup_{i \in I} O_i$. If $x \in \mathcal{O}$, then there exists $i_0 \in I$ such that $x \in O_{i_0}$. Since O_{i_0} is open, there exists $\varepsilon > 0$ such that $B(x, \varepsilon) \subset O_{i_0} \subset \mathcal{O}$. Thus, $\mathcal{O}$ is open.
2. Assume I is finite and non-empty. Let $\mathcal{O} = \bigcap_{i \in I} O_i$. Let $u \in \mathcal{O}$. Then for all $i \in I$, $u \in O_i$. Since each O_i is open, there exists $\varepsilon_i > 0$ such that $B(u, \varepsilon_i) \subset O_i$. Because I is finite, $\varepsilon = \min_{i \in I} \varepsilon_i > 0$. Then $B(u, \varepsilon) \subset \bigcap_{i \in I} B(u, \varepsilon_i) \subset \mathcal{O}$. Thus, $\mathcal{O}$ is open.

The properties for closed sets follow symmetrically. $\square$

3.3 Sequential Characterizations

Theorem 9: Sequential Characterization of Closed Subsets

For $F \subset E$, the following assertions are equivalent:

1. F is closed.
2. For every sequence $(x_n)_{n \in \mathbb{N}}$ in F converging in E , its limit $\lim x_n$ belongs to F .

PROOF. $(1 \Rightarrow 2)$: Assume F is closed. We proceed by contradiction. Assume there exists a sequence $(x_n)_{n \in \mathbb{N}}$ in F such that $x_n \rightarrow x_\infty \in E$, but $x_\infty \notin F$. Then $x_\infty \in F^c$. Since F is closed, F^c is open. Therefore, there exists $\varepsilon > 0$ such that $B(x_\infty, \varepsilon) \subset F^c$. Because $x_n \rightarrow x_\infty$, there exists N_ε such that for all $n \geq N_\varepsilon$, $x_n \in B(x_\infty, \varepsilon) \subset F^c$. This implies $x_n \in F \cap F^c = \emptyset$, which is a contradiction. Thus, $x_\infty \in F$.

$(\neg 1 \Rightarrow \neg 2)$: Assume F is not closed. Then F^c is not open. This means there exists $u^* \in F^c$ such that for all $\varepsilon > 0$, $B(u^*, \varepsilon) \not\subset F^c$, which is equivalent to saying $B(u^*, \varepsilon) \cap F \neq \emptyset$. For each $n \geq 1$, choose $x_n \in F \cap B(u^*, \frac{1}{n})$. This constructs a sequence $(x_n)_{n \in \mathbb{N}}$ in F . Since $d(x_n, u^*) < \frac{1}{n}$, we have $x_n \rightarrow u^*$ as $n \rightarrow \infty$. However, $u^* \notin F$, negating the second assertion. $\square$

Theorem 10: Sequential Characterization of Open Subsets

For $U \subset E$, the following are equivalent:

1. U is open.
2. For every $u \in U$ and for every sequence $(x_n)_{n \in \mathbb{N}}$ converging to u in E , there exists $N \in \mathbb{N}$ such that for all $n \geq N$, $x_n \in U$.

3.4 Connectedness and Interior/Closure

Proposition 11: Clopen Subsets

If $X \subset E$ (or $\mathbb{R}$) is both closed and open, then $X = \emptyset$ or $X = E$.

PROOF. Assume for contradiction that $\emptyset \subsetneq X \subsetneq E$. Then there exists $x \in X$ and $y \in X^c$. Define $I = \{t \in [0, 1] \mid x + t(y - x) \in X\}$. Since $x \in X$, $0 \in I$. Since $y \notin X$, $1 \notin I$. Let $z = \sup I$. Clearly, $z \leq 1$. By definition of supremum, there exists a sequence $(t_n)_{n \in \mathbb{N}}$ in I such that $t_n \rightarrow z$. The sequence $x_{t_n} = x + t_n(y - x)$ lies in X . Since X is closed, the limit $x_z = x + z(y - x)$ must belong to X . Thus, $z \in I$, which implies $z < 1$ (since $1 \notin I$). Because X is open and $x_z \in X$, there exists $\varepsilon > 0$ such that $B(x_z, \varepsilon) \subset X$. This means for sufficiently small $\delta > 0$, $x_{z+\delta} \in X$, implying $z + \delta \in I$. This contradicts $z = \sup I$. Therefore, $X = \emptyset$ or $X = E$. $\square$

Definition 12: Interior and Closure

Let $X \subset E$.

- An element $x \in X$ is an **interior point** of X if there exists $\varepsilon > 0$ such that $B(x, \varepsilon) \subset X$.
- The **interior** of X , denoted $\overset{\circ}{X}$, is the set of all interior points of X . It is the largest open subset contained in X , given by $\overset{\circ}{X} = \bigcup_{O \subset X, O \text{ open}} O$.
- An element $x \in E$ is **adherent** to X if for every $\varepsilon > 0$, $B(x, \varepsilon) \cap X \neq \emptyset$.
- The **topological closure** of X , denoted $\overline{X}$, is the set of all adherent points to X . It is the smallest closed subset containing X , given by $\overline{X} = \bigcap_{F \supset X, F \text{ closed}} F$.